

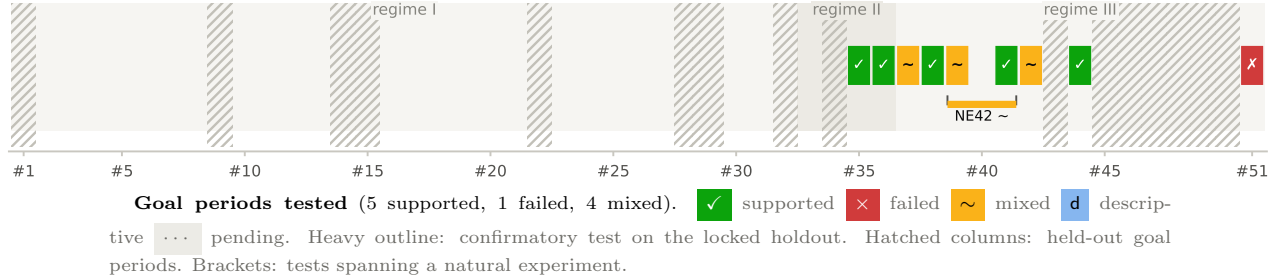
H47: Rooms set the coherence length

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

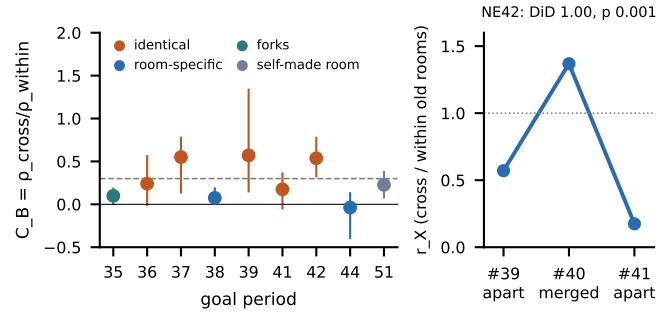
Bottom line. Rooms bound coherence when they do different things, and the bound moves with the chat channel; inside rooms coherence follows conversation, and no room leads.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

The degrees of freedom are the agents' 32-d content vectors in 30-min windows. The test compares per-pair content correlation within rooms (ρ_w) and across rooms (ρ_c) on 9 multi-room periods, against room-relabel nulls. Three native tests follow: the NE42 merge and split, #41's identical instructions and #51's #focus room. Synthetic worlds come first. Rooms bound coherence: $C_B = \rho_c / \rho_w \leq 0.3$ in 6/9 periods (median 0.23). The merge of #best and #rest joins their content fluctuations, and the split restores the boundary (DiD 1.0, p 0.001). Inside rooms, the correlation sits in pairs that talk. Post hoc, the boundary sharpness tracks topic separation. A per-room detector gives no gain (2 clean events). No room leads: both rooms switch within minutes. **Mixed.** The holdout script exists and is not run.



Observables. Left: room contrast C_B per period (30-min windows, day-bootstrap CI). Boundaries are sharp where rooms get different instructions (#38, #44, blue) or separate game forks (#35). They are weak where both rooms get the same kickoff (orange: #37, #39, #42). #41 is the exception: its rooms pick different topics on day 1. Right: NE42. The cross-room correlation, relative to within-room pairs, rises to 1.37 in the merged week and drops to 0.17 after the split.

PHYSICAL MODEL

Soft-spin $O(n)$ vector field (model 11) on a room graph. The spin s_i is an agent's 32-d whitened statement mean per 30-min window, minus its unit mean and static goal directions. Distance is categorical: same room, different room and, inside a room, high and low mention tiers. Fields are global (goal, schedule) or room-specific (instructions, tasks). The coupling runs through the room's chat. S_i is the split-half signal variance.

$$\rho_{w,c} = \frac{\sum_{\text{pairs}} \langle \delta s_i \cdot \delta s_j \rangle}{\sum_{\text{pairs}} \sqrt{S_i S_j}}, \quad C_B = \frac{\rho_c}{\rho_w}, \quad G = \frac{\rho_{w,\text{low}}}{\rho_{w,\text{high}}}$$

$$r_X = \frac{\rho(\text{cross old rooms})}{\rho(\text{within old rooms})}, \quad L = \langle y_{\text{best}}(\tau) - y_{\text{rest}}(\tau) \rangle_{\tau < 2h}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To keep a group's content independent, give it its own room and a different task. Two rooms with the same task share about half of each room's content fluctuation. Rooms with different tasks share almost none. A merge joins coherence within days, and a split restores the boundary at once (NE42). Inside a 25-agent room, coherence sits in conversation pairs, so a monitor must track pairs, not the room mean. Goal changes reach every room within minutes, so no leader room exists to watch. A per-room topic alarm adds nothing to H36's swarm alarm. C_B cannot separate room coupling from room-specific task fields, so it gives no handle for steering through coupling.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 $C_B \leq 0.3$, $p < 0.05$ in $\geq 2/3$	6/9 (3 mixed); median 0.23	passed
P2 flat inside rooms ($G \geq 0.6$)	median G 0.40; #51 one room 0.18	failed (half)
P3 per-room R1 beats swarm R1	AUC 0.62 vs 0.79; clean events: none fire	failed
P4 no room leads	1/8 sig. (#38, a level offset); Stouffer p 0.38	passed
P5 #41 sharp with same instructions	C_B 0.17, p 0.003	passed
P6 instruction direction carries $\leq 15\%$	-0.02 , -0.01 (#38, #44)	passed
P7 NE42 boundary follows channel	DiD 1.00 [0.50, 1.34], p 0.001; #39 above 0.3	mixed (DiD met)
P8 #focus cuts its members	already decoupled before (0.08 / 0.27 / 1.33)	failed
Post hoc: C_B vs separation	Spearman -0.83 (p 0.014, 8 peers)	post hoc

GOAL PERIODS IN DETAIL

G35	✓ supported
G36	✓ supported
G37	~ mixed
G38	✓ supported
G39	~ mixed
G41	✓ supported
G42	~ mixed
G44	✓ supported
G51	× failed
NE42	~ mixed

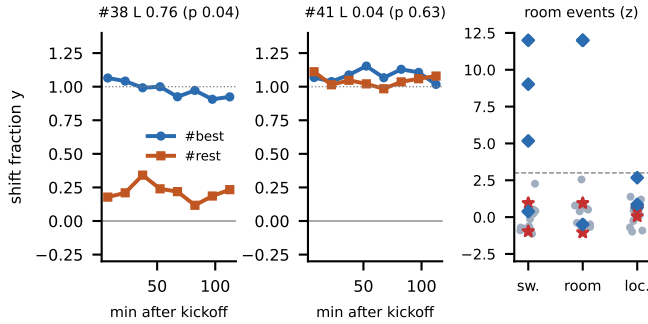
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

C_B is the global share of a room’s fluctuation. Room coupling and room-specific task fields both lower it, and the synthetic worlds show that C_B cannot separate them. Mention tiers are endogenous: co-workers both talk and share topics. NE42 is one goal-confounded A-B-A with 15 agents. The detector test has 2 clean events and 11 placebo days. The quench completes in 1–3 min, below the 15-min bins. Units are 1–17 days long, and P1 passes exactly at its threshold. The post hoc patterns are not tested. The test uses one embedding model; style residuals are checked.



Second figure. Left, middle: kickoff response (shift fraction y , 15-min bins). Both rooms switch in the first bin. The significant lead index of #38 is a level offset, not a lag. Right: room-event scores (stars: clean; diamonds: goal-confounded) against multi-room placebo days (gray), for the swarm, per-room and localized detectors. Only the goal-confounded events fire.

ROUND 2

Test coupling directly with the DQ1 context ledger: does an agent’s content move toward the room messages that it read? Then run the confirm script (#45–#47, #50, NE15) after the reuse disclosures.

- **Essence.** where the boundaries of shared content are in an agent swarm, and whether they are set by the channel (who can read whom), the task, or conversation. Round 1: the channel bounds it (NE42), the task sets how sharp the boundary is, and conversation shapes it inside a room.
- **H47-R1** Read-out-conditioned coupling (the mechanism test).
- **H47-R2** Coherence length inside rooms on the reply graph.
- **H47-R3** Confirmatory run (confirm.py: #45–#47, #50, NE15) after the reuse disclosures, and coordination with H26’s confirm (shared targets).
- **H47-R4** Instrument: DQ5’s gte embedding swap and the final style-residualized vectors.
- **H47-R5** Drop or redesign leadership.
- **H47-R6** Detector: per-room R1 can only be evaluated on more room events; the onboarding and showcase rooms inside the holdout windows (#46, #50) are the remaining ones.

H47-room-coherence-length · generated 2026-10-04 11:51 by infra/summaries/build_summaries.py · card hypotheses/H47-room-coherence-length/README.md · ratings: H47 agent (suggested) (page 2)